Speech Synthesis Evaluation

Speaker Notes — Slides 9–15

Estimated delivery time: ~3–3 minutes | Total words: ~393

Slide 9: Speech Synthesis Overview and Tools

Speech synthesis converts text to audio in five steps: text analysis, linguistic processing, phoneme generation, prosody modeling, and waveform output. We evaluated three free tools: gTTS (Google), edge-tts (Microsoft), and MMS-TTS (Meta). gTTS and edge-tts are cloud-based; MMS-TTS runs offline. We chose 15 hard-to-pronounce sentences, five per language, to expose real quality differences.

~63 words

Slide 10: Methodology

We used a round-trip approach: synthesize a sentence, transcribe it with faster-whisper, then score word by word against the original. Bangla required a Bengali-specific Whisper model because standard Whisper outputs Devanagari script, giving 0% scores despite correct audio. All 45 results are stored in master_results.json.

~54 words

Slide 11: Test Sentences

The 15 sentences were chosen for difficulty: five English tongue-twisters like "She sells seashells", five Hindi sentences with retroflex sounds (e.g., kachcha papara pakka papara), and five Bangla sentences with nasalized and rare vowel sounds. Easy sentences cannot separate tools, so difficulty was intentional.

~54 words

Slide 12: Accuracy Results, English

gTTS led at 97.1%, edge-tts at 76.1%, MMS-TTS at 57.4%. All three scored 100% on E1. The hardest was E3, averaging 61.9% across tools. MMS-TTS scored 0% on Peter Piper, and edge-tts returned digits instead of words on E3. Cloud tools clearly outperformed offline.

~51 words

Slide 13: Accuracy Results, Hindi

Hindi was harder overall. gTTS and edge-tts tied at 57.6%; MMS-TTS trailed at 37.0%. The toughest sentence, kachcha papara, scored 50% for cloud tools and 0% for MMS-TTS, because tools mistook the retroflex consonant for a plain r. The offline model was weakest.

~51 words

Slide 14: Accuracy Results, Bangla

All three tools clustered tightly between 72.8% and 73.3%, making Bangla the most balanced language. The hardest sentence scored 57.1% across all tools due to the letter khanda ta being rendered differently by each engine. B5 with the rare vowel ri was the easiest, with gTTS and MMS-TTS both at 100%.

~57 words

Slide 15: Notable Errors and Tool Comparison

Key errors: MMS-TTS failed Peter Piper (0%); cloud tools misread the Hindi retroflex as r; all engines disagreed on Bangla's khanda ta. Summary: gTTS is best for English at 97.1%; cloud tools tie on Hindi and Bangla; MMS-TTS is the only offline option but less accurate. Five sentences per language is indicative, not conclusive. Next: machine translation.

~63 words